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Enhanced Control Design of Simple DC-DC Boost Converter-driven DC Motors: Analysis and Implementation

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Abstract—The DC-DC boost converter is one of the simplest power electronic devices that has not been yet exploited in a wide range of industrial applications due to control design difficulties caused by its model inherent special structure. Such an industrial application is the DC motor speed regulation that is studied in the present work. Particularly, in this article, a novel, non-linear control scheme for the duty ratio input of the converter is proposed, which is extensively analyzed and experimentally tested. The proposed design, though non-linear, results in a very simple scheme, ensures that the duty ratio takes values exclusively in the permitted range $[0, 1)$, achieves precise speed regulation even in cases of high unknown load disturbances, and does not depend on system parameters and states. Simultaneously, the design is formulated in a manner that provides a closed-loop passive system, which, as proven in the article, satisfies all these assumptions and properties that make possible the application of a new advanced non-linear method that strongly connects passivity with stability. Thus, the boundedness of all the closed-loop states and the stability and convergence to the desired steady-state equilibrium are directly concluded. The theoretical analysis is verified through extended simulation and experimental results.

1. INTRODUCTION

DC motors are still used in many industrial applications that require precise speed or torque control, such as electric rail traction, electric cranes, robotic manipulators, etc. [1–5]. To achieve the desired regulation, currently, electrical power converters that provide the desired DC voltage are used [6–9]. In low-medium-power applications, the most common, simple, and low-cost DC-DC converter is the boost converter. Thus, a simple DC-drive scheme can include a DC-DC boost converter feeding a separately excited or permanent magnet DC motor. Since the DC-DC boost converter is used to boost the input voltage to a higher level, this scheme can be used in applications where precise motor speed regulation is demanded above a minimum of speed, corresponding to the lower output voltage, which is practically

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equal to the input voltage [10, 11]. Hence, the control task is to provide effectively the duty ratio switching function to achieve a stable and precise convergence to the desired equilibrium. Usually, the approach for obtaining a high-standard control scheme is of the model-based type. Consequently, the controller design requires the complete converter and DC motor modeling. As it has been shown in the literature, for control purposes, the average converter model combined with the separately excited DC motor model is adequate [10, 12]. Unfortunately, due to the non-linear dynamic converter model, the main advantage of the separately excited DC motor that is its linear description is lost; furthermore, the complete model is in a special non-linear form [10].

Opposite to its circuit simplicity, the control design of the DC-DC boost converter duty ratio input that ensures stability and convergence to equilibrium seems to be a difficult task, especially for cases where a dynamic system is being fed [11]. This constitutes a challenging problem because inherent system difficulties, such as the unstable zero dynamics, put a hard constraint in the control design, which is well-known in the power and industrial electronics community [13]. A common solution to this problem has been given by regulating indirectly the output voltage via the input inductor current. In this frame, different control techniques have been applied for the regulation of a DC-DC boost converter at a desired output voltage [14, 15]. Early designs propose either simple linear lead-lag compensators or conventional, simple, or cascaded proportional-integral (PI) controllers, which are based on small-signal model analysis and linearization [16, 17]. These types of controllers are very sensitive to model and load uncertainties since the changes of the operating point reduce the small-signal model validation. Several researchers have tried to improve the converter performance by proposing intelligent techniques (fuzzy, adaptive fuzzy) [18–20] or non-linear schemes [21], such as sliding-mode control [22–25]. However, fuzzy and sliding-mode controllers, especially when sliding surfaces are dependent from the operating point, cannot guarantee that the closed-loop system will remain stable after large input variations or load disturbances. Additionally, it is noticed that in most cases, all the previous designs are referred to simple non-dynamic loads where a simple output voltage regulation is proposed [26–28]. Other non-linear control techniques, such as the passivity-based control (PBC) technique, have some design advantages but require the error dynamics model of the complete system, which is different for every dynamic load, and it is dependent from the system parameters and the operating point. Sometimes, adaptive control techniques are added to enhance the PBC performance, which complicates the entire system [29, 30]. In PBC design, the most advanced technique pro-

posed is the interconnection and damping assignment (IDA-PBC), which guarantees closed-loop system stability of the non-linear model [31]. However, it is worth noting that PBC techniques require the resolution of a system of either partial differential equations or analytic equations, which, however, depend on the system parameters and cannot be solved in cases where a dynamic load is considered, such as a DC motor.

The case of driving a DC motor by the use of a boost converter is referenced in a very limited number of recent works. This is due to the fact that the industrial interest is low compared with other solutions, such as that of using full AC/DC converters, while increased difficulty of the closed-loop system analysis has arisen by the non-linearities introduced by the boost converter model. In this frame, the different control laws that have been proposed include cascaded PI or hysteresis current combined with PI controllers [32]. However, they remain less robust, especially when applied under natural circumstances [14, 33–35]. Fuzzy and neuro-fuzzy control designs have been also proposed for boost converter driven DC motors, but they still suffer by the absence of adequate stability analysis [36–38] with their performance to be evaluated by a rather empirical manner. In non-linear designs, passivity-based techniques seems to be more effective since they are based on the exact non-linear model of the total converter-motor system, but they highly depend on the parameters of the model [10, 11, 14].

In this article, a non-linear dynamic controller is proposed and applied on a DC-DC boost converter driving a separately excited DC motor. The aim of the controller design is to regulate the motor speed by providing directly the duty ratio signal input of the boost converter, bounded in the permitted range without needing external limiters. To this end, the performance of the system is analyzed by using its entirely converter-motor non-linear model. The controller performance is established through a general form of dynamic equations, which are taken into account for a complete stability analysis. The proposed duty ratio controller, though non-linear, results in a very simple implementation, while it does not require the knowledge of the system parameters or the load torque, and the only measured state variable is the one to be regulated (motor speed). It is clear that the proposed design does not belong to the indirect regulation methods, and therefore, it does not need a cascaded structure or an adaptation mechanism. A simple mathematical model describes the controller operation in a manner that permits the controller states to move on the circumference of a specific circle until the resulting duty ratio converges to the equilibrium that corresponds exactly to the desired speed regulation. Simultaneously, this structure guarantees boundedness of the duty ratio signal in a permitted range $[0, (1 - \delta)^2] \in [0, 1)$, since δ is arbitrarily selected as

a very small constant. As the DC-DC boost converter has by construction a lower bound of the output voltage, the duty ratio is initially set to zero, which provides a corresponding lower bound of the motor speed ω_0 . As the controller states reach the desired circumference, the proposed scheme is activated, resulting thereafter in a stable and bounded performance. As it is proven in the article, the closed-loop system satisfies the passivity property in a way that ensures the existence of damping characteristics of the entire system. Thus, the analysis presented lies in the field of designing stable controllers based on the passivity property of the system. However, the method has been developed on a new theoretical basis [39, 40] that exploits the Hamiltonian-passive structure of the system and is valid for unknown constant or piecewise constant external inputs and/or disturbances. Following this approach, it is easily concluded that the closed-loop system meets all the assumptions and requirements presented by the authors in [39], and therefore, stability and convergence to the desired steady-state equilibrium is guaranteed. The method used is clearly different from the PBC technique, making possible to prove stability independently from the reference signal changes and the load torque disturbances (of course between certain limits according to the ratings of the converter and motor). The analysis is verified through extended simulation results, and the controller performance is also experimentally evaluated on a suitable experimental setup wherein the results completely match the theoretical analysis and simulations.

Only recently, a similar method has been established [41] for the specific case of a PI controller applied on a series-excited DC motor. In that study, the method is developed to include the P and I terms, while in the present article, a general non-linear dynamic controller is proposed. More specifically, the duty ratio control law is provided by a quadratic function of the controller dynamics. This simplifies the analysis and ensures the required boundedness of the control signal without needing any other constraint, as clearly imposed in [41]. Therefore, in this case, the general case of a separately excited DC motor is addressed. It is noted, however, that in both cases a similar stability analysis is used.

The article is organized as follows. In Section 2, the dynamic model of a DC-DC boost converter connected to a separately excited DC motor is presented, and its unique equilibrium is derived. In Section 3, the new non-linear dynamic speed controller is proposed, and the closed-loop system is analyzed. Furthermore, stability and convergence to the desired equilibrium for the closed-loop system are proven by using an extended analysis. In Section 4, the experimental setup of a boost converter driven DC motor, used to verify the effectiveness of the proposed approach, is presented. Simulation and experimental results that indicate the application of the

proposed non-linear controller for several changes of the reference signal or load torque step disturbances are shown in Section 5. Finally, in Section 6, some conclusions are drawn. The Appendix presents several model material required for the theoretical analysis.

2. MODEL OF BOOST CONVERTER AND SEPARATELY EXCITED DC MOTOR

It is evident that many industrial electronic applications have power electronic devices as fundamental elements because of their ability for processing power in a more efficient way. In the present case, a simple DC-DC boost converter is used to drive a DC motor at a specified rotor speed. Figure 1 shows a separately excited DC motor driven by a DC-DC boost converter. To analyze the system performance and determine an efficient drive, the exact system model is needed. Thus, taking into account the DC motor dynamics [4] as well as the average value model of the boost converter [12], adequate for control purposes, and applying Kirchhoff's laws, the complete model of the system is obtained as follows:

$$\begin{aligned} L\dot{i} &= -(1-\mu)v + E, \\ C\dot{v} &= (1-\mu)i - i_a, \\ L_a\dot{i}_a &= v - R_a i_a - K_e \omega, \\ J\dot{\omega} &= K_e i_a - b\omega - T_L, \end{aligned} \quad (1)$$

where i is the inductor current, v the capacitor voltage, i_a the armature current, and ω the angular velocity (motor speed). Referring to the DC-DC converter, L and C are the converter inductance and capacitance, respectively, E is the DC input voltage, and μ is the duty ratio representing the switching cycle ratio in a period. For the DC motor, R_a and L_a are the armature resistance and inductance, respectively, K_e is the motor constant, J the motor inertia, b the friction coefficient, and T_L the load torque. The magnetic saturation effect is not taken into account, while the field winding is assumed to be fed by a constant external voltage V_f (Figure 1). The DC motor is directly connected to the output of the boost converter; it should be noticed that this is the worst system representation since no parasitic conductance or resistance has been added in parallel to the capacitor or in series to the input inductor, respectively. As it is well known, it is misleading to add these dissipative elements, since they falsely increase the system damping and stability [14].

In the system in Eq. (1), the control input is the duty ratio μ of the converter, which should be bounded in the region $[0, 1]$. Additionally, the converter input voltage E and load torque T_L represent the uncontrolled external inputs. Often, E is constant while T_L can be considered as a fully unknown, piecewise constant input.

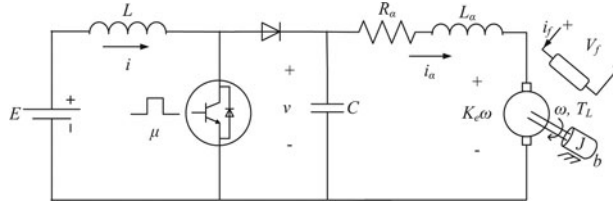


FIGURE 1. Model of a boost converter driven DC motor.

By setting $\dot{x} = 0$ in Eq. (1), where $x = [i \ v \ i_a \ \omega]^T$, one can derive the unique equilibrium point [14] of the converter-motor system. Assuming a specific duty ratio μ^* and a given input voltage E and load torque T_L , the equilibrium of the system in Eq. (1) is

$$(i^*, \ v^*, \ i_a^*, \ \omega^*) = \left(\frac{b(K_e E - R_a(1 - \mu^*)T_L)}{K_e(1 - \mu^*)^2(R_a b + K_e^2)} + \frac{T_L}{(1 - \mu^*)K_e}, \ \frac{E}{1 - \mu^*}, \frac{b(K_e E - R_a(1 - \mu^*)T_L)}{K_e(1 - \mu^*)(R_a b + K_e^2)} + \frac{T_L}{K_e}, \frac{K_e E - R_a(1 - \mu^*)T_L}{(1 - \mu^*)(R_a b + K_e^2)} \right). \quad (2)$$

It is remarkable that if the duty ratio reaches the upper limit, *i.e.*, $\mu = 1$, the system in Eq. (1) leads to instability since the inductor current i tends to infinity. Hence, it becomes obvious that really $\mu \in [0, 1)$; indeed, in practical applications, the upper value of μ is much less than 1.

3. NON-LINEAR CONTROLLER DESIGN AND ANALYSIS

3.1. A Simple Stable Non-linear Speed Controller

In accordance with the system model given in the previous section, the controller aim is to effectively regulate the control signal μ of the power converter in a way that drives the motor speed ω from an initial condition to a desired value ω^* and to maintain the speed constant independently from any torque unknown disturbances.

In this article, the main purpose is to illustrate that it is possible to design simple stable controller structures, which directly provide the converter duty ratio μ exclusively in the range $[0, 1)$, where the pulse-width modulation (PWM) technique can effectively drive a DC motor at a desired speed. It is important to notice that since the DC-DC boost converter can only boost the input voltage E to a higher level ($v \geq E$), then, as happens in most control applications [12], the controller is activated when the system is at an initial equilibrium with minimum duty ratio $\mu_0 = 0$. From 0 to ω_o , the system operates as an open-loop system, with the duty ratio set at $\mu = 0$, corresponding to an armature voltage (converter output voltage)

equal to input voltage level E and until the motor reaches a speed ω_o , as is defined by the equilibrium given by Eq. (2) for $\mu = 0$. In this case, the system is linear time invariant with stable eigenvalues of the plant matrix assigned in the left half-plane. Therefore, the motor will always effectively reach a low initial speed ω_o , thus making possible the practical implementation of the controller in advanced applications. Taking these concepts into account, the following simple non-linear controller for the duty-ratio μ is proposed:

$$\mu = \begin{cases} 0 & \text{for } 0 \leq t < T_0 \\ p^2 & \text{for } t \geq T_0 \end{cases} \quad (3)$$

with dynamics

$$\begin{bmatrix} \dot{p} \\ \dot{q} \end{bmatrix} = \begin{bmatrix} 0 & k(\omega^* - \omega) \\ -k(\omega^* - \omega) & -c(p^2 + q^2 - c_0^2) \end{bmatrix} \begin{bmatrix} p \\ q \end{bmatrix}, \quad (4)$$

where p and q are the control states, k and c are positive constant gains, and $c_0 = 1 - \delta$ is defined with δ as an arbitrarily small positive constant and T_0 the first time instant at which $p^2 + q^2 \rightarrow c_0^2$ and $\omega \rightarrow \omega_o$, thus enabling the proposed controller activation in accordance to Eq. (3), for ω^* greater than ω_o .

Notice that the controller design introduces a very simple structure and is fully independent from the system parameters, and the load torque and requires only the motor speed measurement. Thus, the simple implementation of the proposed controller, useful for practical applications, is shown in Figure 2. It is also noted that opposite to the traditional boost converter fed applications, wherein the measurement of the input current is essential for controlling the desired state variable (voltage or speed) [10–12, 14, 17, 21], in the proposed scheme, the current sensor is not required, and the non-linear stability is maintained, further demonstrating the superiority of the proposed method. Additionally, it should be clarified that the machine and converter protection that can be provided by the current controller no longer exists. Nevertheless, in this case, the guaranteed bounded controller performance compensates for this feature. In general, the proposed non-linear controller represents a simple and efficient solution that can ensure stability and inherent duty ratio boundedness in the permitted range (without needing external saturators), as the following proposition states.

Proposition 1. *The duty-ratio controller in Eqs. (3) and (4), with at least one initial condition $p(0)$ or $q(0)$ arbitrarily selected at any non-zero constant value, provides a duty ratio μ constrained in the permitted range $[0, (1 - \delta)^2]$ and guarantees stability for all $t \geq 0$ and convergence to the steady-state equilibrium of the system in Eq. (1) corresponding to the desired motor speed ω^* .*

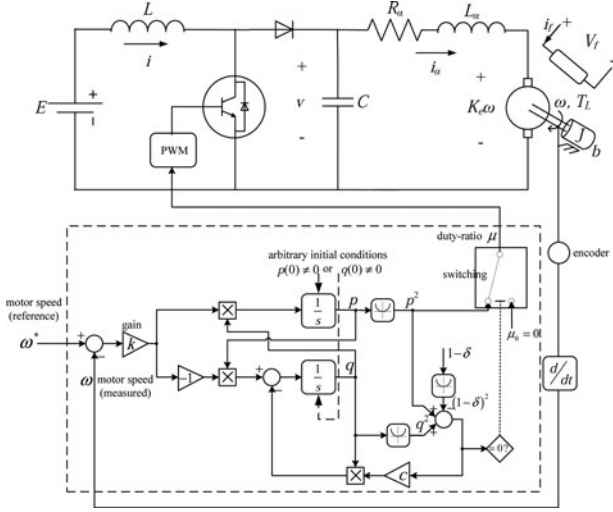


FIGURE 2. Schematic diagram of the proposed control implementation.

Next, an outline of the proof of Proposition 1 and a discussion is given.

As shown in Figure 2, initially, the duty ratio μ is set equal to zero and the motor speed converges to ω_0 . After $t = 0$, considering arbitrary initial conditions for p and q as described by Proposition 1, and considering the following Lyapunov function for the controller dynamics in Eq. (4)

$$W(t) = p^2(t) + q^2(t) \quad (5)$$

the time derivative of $W(t)$ is obtained as

$$\begin{aligned} \dot{W}(t) &= 2k(\omega^* - \omega_0)p(t)q(t) - 2k(\omega^* - \omega_0)p(t)q(t) \\ &\quad - c(p^2(t) + q^2(t) - c_0^2)q^2(t) \\ &= -c(p^2(t) + q^2(t) - c_0^2)q^2(t). \end{aligned} \quad (6)$$

It can be seen from Eq. (6) that $\dot{W}(t) > 0$ when $p^2 + q^2 < c_0^2$ and $\dot{W}(t) < 0$ when $p^2 + q^2 > c_0^2$, while $\dot{W} = 0$ only when

$$p^2 + q^2 = c_0^2. \quad (7)$$

Note that the case $q = 0$, which also leads to $\dot{W}(t) = 0$, is clearly covered by Eq. (7).

This implies that the circle with its center the origin and radius equal to $c_0 = 1 - \delta$ attracts the control states p and q on its circumference, *i.e.*, acts as an attractive circle on the p - q plane (Figure 3). Hence, starting from any arbitrary initial point inside (point A) or outside (point A') of the circle, there definitely exists a time instant T_0 when p and q reach the circumference of the circle Eq. (7) and speed converges to ω_o . Practically, it is considered that p and q reach the circumference when $|p^2 + q^2 - c_0^2| < \gamma$, where γ is an arbitrary very small positive constant. This enables the controller to be activated for $t \geq T_0$, as shown in Figure 2. Note that parameter c describes

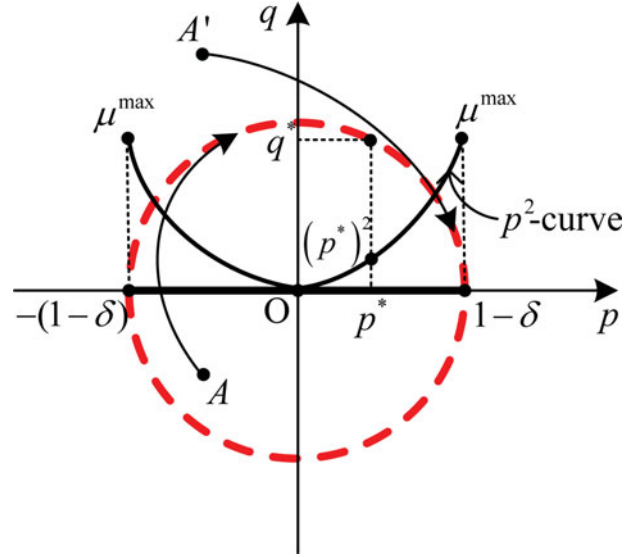


FIGURE 3. Attractiveness of the control states on a circle with radius $1 - \delta$.

the rate of attractiveness on the circle, *i.e.*, for large c , the control states p and q quickly reach the desired circumference.

As a result, for $t \geq T_0$, it holds true that $p^2(t) + q^2(t) = c_0^2$, since once the states reach the desired circle, they cannot leave for any future time (see [42]). At this point it is significant to observe that after T_0 , the diagonal term $-c(p^2 + q^2 - c_0^2)$ of the controller dynamics in Eq. (4) can be omitted. Thus, without loss of generality, after T_0 , the controller dynamics, namely p and q as given by Eq. (4), are moving exclusively on the circumference of the predetermined circle with angular velocity depending on the difference between the desired ω^* and the measured ω motor speed, respectively. Therefore, it becomes obvious that p can take any value in the range $[-c_0, c_0] = [-(1 - \delta), 1 - \delta]$, and as a result, the duty ratio μ , as given by Eq. (3), is restricted in the range $[0, c_0^2] = [0, (1 - \delta)^2]$. Figure 3 shows the range wherein p moves, while one can easily see from the p^2 -curve that indeed μ takes values from 0 to a maximum $\mu^{\max} = (1 - \delta)^2$. As it is indicated by Figure 3, as ω reaches ω^* , then p and q can reach their equilibriums p^* and q^* , respectively, while the desired duty ratio value is given by $\mu^* = (p^*)^2$ on the p^2 -curve, definitely found in the permitted range $[0, (1 - \delta)^2]$. As a result, it is proven that μ lies in a permitted bounded operative area in which the desired μ^* exists.

Now, it remains to prove stability and convergence to the aforementioned equilibrium. To this end, the complete closed-loop system should be analyzed.

The stability analysis that is briefly described in the following is based on a new method, recently presented in [39, 40]. This new method strongly relates the closed-loop passiv-

ity property of a certain class of Hamiltonian-passive systems to the stability and convergence to equilibrium. Although in the present case the control law in Eq. (3) and the controller dynamics in Eq. (4) are quite different from those analyzed in [39], one can easily see that after proving the closed-loop passivity property, a similar stability analysis can be effective.

The idea of applying the passivity concept in industrial electronics applications is not new, as pointed out in [12, 14]. Passivity characterizes the dynamical systems in which the energy exchanged with the environment plays a central role. In passive systems, the rate at which the energy flows into the system is not less than the increase in storage. Roughly speaking, a passive system cannot store more energy than that supplied to it from outside, with the difference being the dissipated energy [43]. Given this energy interpretation, it is clear that passivity is intimately related with the physics of the system and, in particular, with its stability properties. These fundamental properties have motivated many researchers to take passivity as the basic building block for stabilizing non-linear systems [40, 43].

In the present study, the contribution is twofold. First, the complete non-linear system (converter-motor) is considered on which the proposed duty ratio controller is applied in a manner that ensures closed-loop passivity. On the other hand, a new passivity-based method, as proposed [39, 40], is used that clearly differs from the conventional ones (such as PBC and others) and that can include simple and effective design solutions that guarantee stability and convergence to equilibrium, independently from the system parameters, load changes, or operating point.

Thus, to proceed with the analysis, it is first proven that the closed-loop system has the passivity property.

Assuming that the control input μ of the system is given by Eqs. (3) and (4) and taking into account Eq. (7), the closed-loop system can be written in the following non-linear form:

$$\tilde{M}\dot{\tilde{x}} = (\tilde{J}(p, q, \omega) - \tilde{R})\tilde{x} + \tilde{\epsilon}, \quad (8)$$

where

$$\begin{aligned} \tilde{x} &= [x^T : p \ q]^T = [i \ v \ i_a \ \omega : p \ q]^T, \\ \tilde{\epsilon} &= [E \ 0 \ 0 \ -T_L \ 0 \ 0]^T, \end{aligned}$$

while matrix $\tilde{M}$ is constant, symmetric, and positive definite, $\tilde{J}$ is skew-symmetric and $\tilde{R}$ is symmetric and positive semi-definite, all given in the Appendix. Also, as discussed in the previous section, $\tilde{\epsilon}$ includes unknown constant or piecewise constant inputs.

For the system in Eq. (8), consider the following storage function $\tilde{H}$ that describes the energy stored:

$$\tilde{H} = \frac{1}{2} \tilde{x}^T \tilde{M} \tilde{x}. \quad (9)$$

Then the time derivative of the storage function $\tilde{H}$ is calculated as

$$\dot{\tilde{H}} = \tilde{x}^T \tilde{M} \dot{\tilde{x}}.$$

Substituting, however, $\dot{\tilde{x}}$ from the closed-loop system dynamics in Eq. (8) yields

$$\begin{aligned} \dot{\tilde{H}} &= \tilde{x}^T ((\tilde{J}(p, q, \omega) - \tilde{R})\tilde{x} + \tilde{\epsilon}) \\ &= \tilde{x}^T \tilde{J}(p, q, \omega)\tilde{x} - \tilde{x}^T \tilde{R}\tilde{x} + \tilde{x}^T \tilde{\epsilon} \\ &= -\tilde{x}^T \tilde{R}\tilde{x} + y^T u, \end{aligned} \quad (10)$$

where $y = [i \ \omega]^T$ and $u = [E \ -T_L]^T$; i.e., Eq. (10) yields

$$\dot{\tilde{H}} \leq y^T u. \quad (11)$$

According to the passivity theorem [43, 44], Eq. (11) proves that the closed-loop system remains passive after the proposed control application. This implies that the energy stored in the system is always less or equal to the energy injected by the external sources, and as Eq. (10) indicates, the closed-loop system includes damping elements (since $\tilde{R} \neq 0$).

The closed-loop system in Eq. (8) is in Hamiltonian-passive form, as described in [39], with all the non-linearities limited into the $\tilde{J}(p, q, \omega)\tilde{x}$ term. Besides the significant difference of the control scheme in Eqs. (3) and (4), with respect to the general controller proposed in [39], it is easily seen that the closed-loop system can be handled in a similar manner. In particular, the control operation provides a bounded duty ratio input in the range $[0, (1 - \delta)^2]$, and according to the previous analysis, it becomes evident that Assumptions 1–3 mentioned in [39] are satisfied. As a result, following the stability analysis described in [39, 40] it can be proven that the proposed controller guarantees closed-loop stability of the entire system and convergence to the equilibrium that corresponds to the desired motor speed ω^* . Thus, not only do states p and q converge to their equilibrium, but the whole state vector $\tilde{x}$, as determined in Eq. (8), converges to its equilibrium $\tilde{x}^*$.

3.2. Effectiveness of the Proposed Control Scheme under Reference Signal Changes and Load Torque Step Disturbances

In the previous subsection, it has been shown that the proposed controller leads to a stable operation and convergence at the desired equilibrium. This can be observed in Figure 4, where the controller states p and q are initialized on an arbitrary point A . According to the previous analysis, at a future time instant T_0 , the states reach the circumference of the circle in Eq. (7)

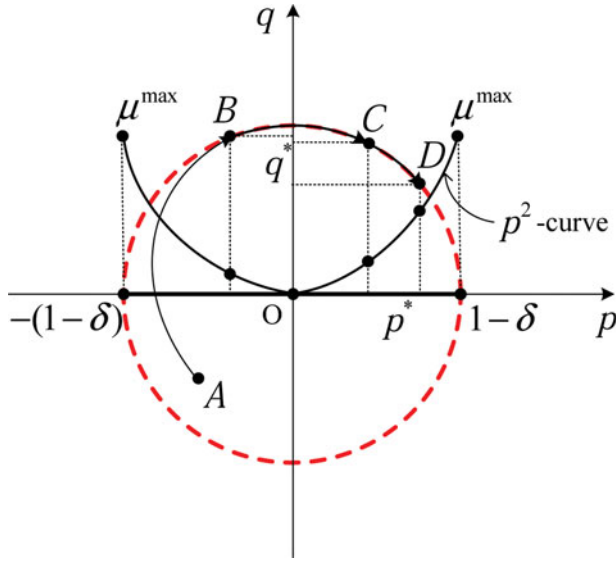


FIGURE 4. Convergence on the desired equilibrium for step changes of the reference signal or step changes of the load torque.

(point B) at an initial equilibrium corresponding to the lower possible motor speed ω_0 , and then the proposed controller is activated by the switching shown in Figure 2. Thus, after T_0 and for any $\omega^* > \omega_0$ states, p and q converge to the new equilibrium at some time $T^- > T_0$ corresponding to the desired ω^* (point C) by moving on the circumference of the circle. Now, after any future step change of the reference signal ω^* , the convergence to any other new equilibrium is guaranteed, for instance, let the case where $\mu \in [0, 1)$ and the state variables $\tilde{x}$ are on an equilibrium at some time $T^- > T_0$, *i.e.*, the states are on an equilibrium $\tilde{x}^*(T^-)$ corresponding to $\omega^*(T^-)$ and the control input is $\mu^*(T^-)$, with control states $p^*(T^-)$ and $q^*(T^-)$ determined by point C on the desired circle as shown in Figure 4, and let a change occur to the reference signal ($\omega^*(T^+) \neq \omega^*(T^-)$). Since the control states are constrained on the circumference of the circle in Eq. (7) for all future time, they definitely reach the new equilibrium values $p^*(T^+)$ and $q^*(T^+)$ corresponding to the new equilibrium described by point D ($\mu^*(T^+)$) (Figure 4) independently from the departure point C .

On the other hand, in most control applications, the load connected to the rotor can perform sudden changes, *i.e.*, in elevators, crane systems, electric rail traction, etc. These changes to the load torque T_L instantly force the motor speed to increase or reduce from the reference speed ω^* , and the question is whether this change leads the system to instability. It is a mandatory requirement for the designed speed control scheme to be able to restore the initial speed after this change of the torque T_L .

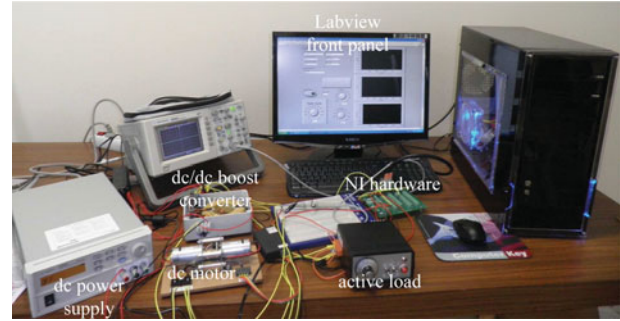


FIGURE 5. Experimental setup.

As stated in Section 2, one can easily see from the open-loop complete model in Eq. (1) that for a certain load torque T_L and a desired speed value ω^* , there exists a unique duty ratio μ^* that corresponds to the desired equilibrium. Assuming now the same reference speed ω^* but a different load torque T'_L , the suitable unique duty ratio $\mu_{T'_L}^*$ differs from the previous one. This corresponds to a new point on the p - q plane, and the controller acts in a similar way as in the case of a reference signal change. Therefore, the same analysis holds true for the control states p and q , as previously described, which move from an initial point, say C , to a new point D , as shown in Figure 4, proving stability and convergence to the desired equilibrium under step load torque changes. Although the effectiveness of the proposed scheme is proven under various changes of the load torque, it should be emphasized that, as in every control scheme, there is a bound on the load torque T_L that a motor can confront at any specific speed. This can be defined by the equilibrium and the parameters of the system in Eq. (1).

4. EXPERIMENTAL SETUP FOR THE BOOST CONVERTER DRIVEN DC MOTOR

The experimental setup depicted in Figure 5 consists of the following devices: a DC power supply for providing the input voltage to the switching converter, a DC/DC boost converter with a driving circuit to command the switching position μ , a separately excited DC motor, a tacho generator operating as a speed encoder, a data acquisition system with a connector block to transfer the data of the motor speed to the PC and send the appropriate pulse to the converter driver, and a computer with Labview program to implement the proposed non-linear controller. A generator is used to produce the external torque by feeding an active load designed in a manner that permits step changes to the load torque.

The data acquisition card PCIe-6321 relies on hardware from National Instruments (NI) while the software used for the control design corresponds to NI Labview (ni.com), which

provides a convenient and functional environment for the user. The motor speed data as taken by the tacho generator are transferred through the PCI card, and a connector block to the computer and the proposed controller generates the appropriate PWM signal (in a frequency of 5 kHz), which is sent from the counter output pin to the converter driver.

The design parameters of the boost converter setup are $E = 12$ V, $C = 57.6$ μ F, and $L = 16$ mH, while the parameters of the DC motor are $P_n = 15$ W, $R_a = 8.9$ Ω , $L_a = 12.5$ mH, $K_e = 50$ mVs/rad, $J = 0.02$ kgm², and $b = 77.613$ μ (Nm $\cdot$ s)/rad.

5. SIMULATION AND EXPERIMENTAL RESULTS

To verify the effectiveness of the proposed control law, the response of the converter-motor system is simulated and tested using the experimental setup.

Three different cases are considered where the control parameters are chosen in all cases to be $k = 0.018$ rad/sec⁻¹ and $\delta = 0.01$, *i.e.*, $\mu \in [0, 0.9801]$ is determined, according to the theoretical analysis. In Case A, the proposed controller is simulated for a simple motor speed regulation to verify the stability analysis and to confirm the performance on a circumference of the proposed controller as described in Section 3. In Case B, several step changes of the reference signal ω^* are simulated and experimentally tested to examine whether the non-linear controller is capable of regulating the motor speed at different speed levels while the load torque is assumed to be constant. Finally, in Case C, the effectiveness of the proposed non-linear controller is examined under various step changes of the load torque T_L by comparing the simulation and experimental responses.

Case A: Motor Speed Regulation

Since the DC motor is driven by a boost converter, the lowest motor speed can be defined when the duty ratio μ is zero, resulting in an output voltage equal to the input voltage 12 V, and according to the system parameters, the rotor speed is approximately initialized at $\omega_0 = 160$ rad/s. The controller task is to regulate the output motor speed to $\omega^* = 200$ rad/s when initially $\omega = \omega_0$ and $T_L = 0.01$ Nm is constant. In Figure 6, the responses of the speed, duty ratio, and controller states are presented. In particular, Figure 6(a) shows a fast time response of the motor speed and regulation at the reference value, while the duty ratio follows a similar response in Figure 6(b). Figure 6(c) shows convergence of the control states p and q at the desired equilibrium, and the theoretical analysis in Section 3 is justified since both states remain on the circumference of the desired circle in the p - q plane as expected and reach the controller equilibrium point B with coordinates $(p^*, q^*) = (0.5686, 0.8104)$.

It is obvious that $(p^*)^2 = 0.3233 = \mu^*$ as expected from the control law in Eq. (3).

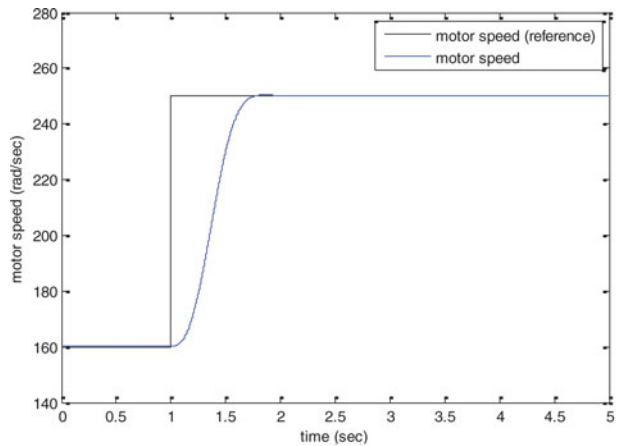
Case B: Effectiveness of the Proposed Scheme under Various Step Changes of the Reference Signal

In this case, the controller performance is tested under various changes of the reference speed in both simulation and experimental environments. The reference signal is initially assumed at 200 rad/sec. At time instant $t_1 = 2$ sec, the reference speed rises to 250 rad/s, at $t_2 = 7$ sec, it drops rapidly to 220 rad/s, and finally at $t_3 = 12$ sec, raises again to 300 rad/s under a constant load torque. Figures 7(a), 7(b), and 7(c) represent the motor speed, armature voltage, and duty ratio responses of the converter-motor system. One can observe that the proposed control scheme regulates the motor speed to any reference level, and the experimental response directly follows the simulated one. The armature voltage and the duty ratio are almost the same; the very small differences that appear between the simulation and the experimental responses are rather due to parameters' deviations [10]. Stability and convergence to equilibrium seems to be held in accordance to the analysis presented in Section 3.

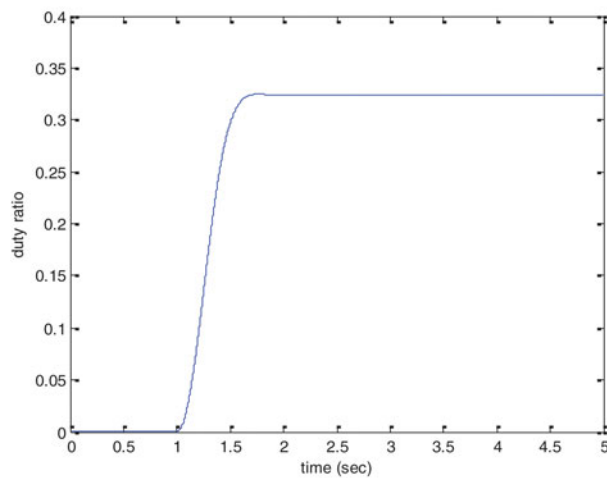
Case C: Effectiveness of the Proposed Scheme under Load Torque Step Disturbances

In this case, two rapid changes of the load torque occur while the system is at a reference speed $\omega^* = 250$ rad/s. At time instant $t_1 = 2$ sec, a 75% increase occurs at the load torque, while at $t_2 = 7$ sec, a 25% drop takes place at the load torque; *i.e.*, the system performance is tested under high torque disturbance changes. From Figure 8, one can clearly see the robustness of the proposed non-linear controller in all load torque changes since the motor speed returns to its reference value after a fast transient. It is remarkable that the experimentally obtained response has its mean value exactly on the simulated one. The step changes of the load torque are experimentally achieved by the use of the DC generator and the active load, as discussed in Section 4.

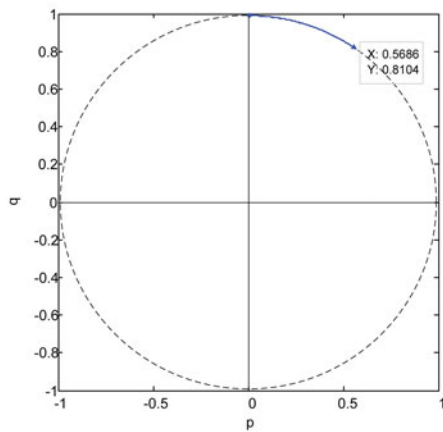
It is noted that in Figures 7(a) and 8, motor speed ripples appear with distinct spikes and dips in the experimental results. These are mainly caused by the torque ripple that appeared in practical implementation of control in power electronic devices. The effects of torque and speed ripples are particularly undesirable in some demanding motion controls. They may lead to oscillations that cause deterioration in the performance. The effective reduction of these ripples and the successful suppression of the spikes and dips need further analysis and constitute an interesting field of research (see, for example, [45]).



(a)

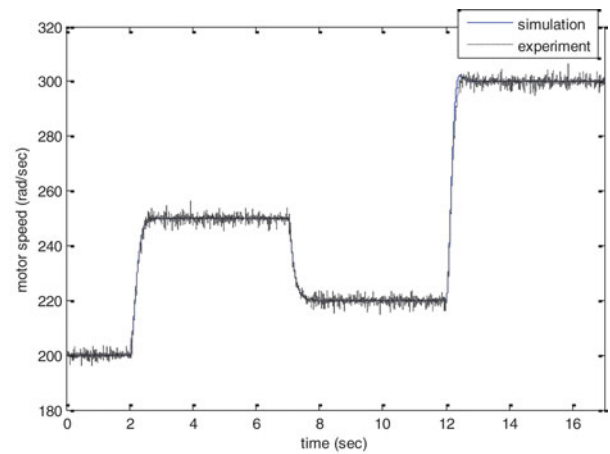


(b)

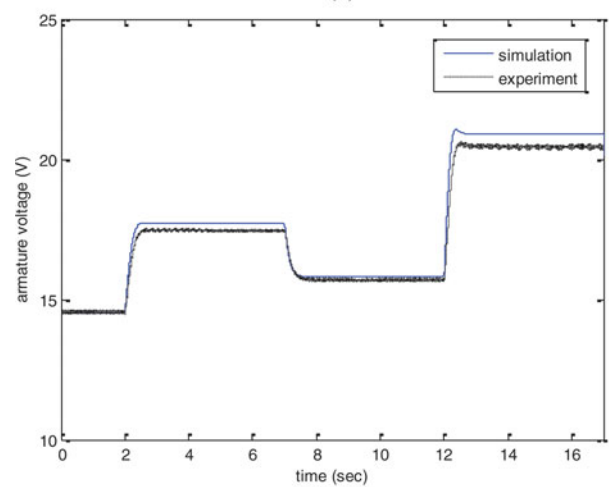


(c)

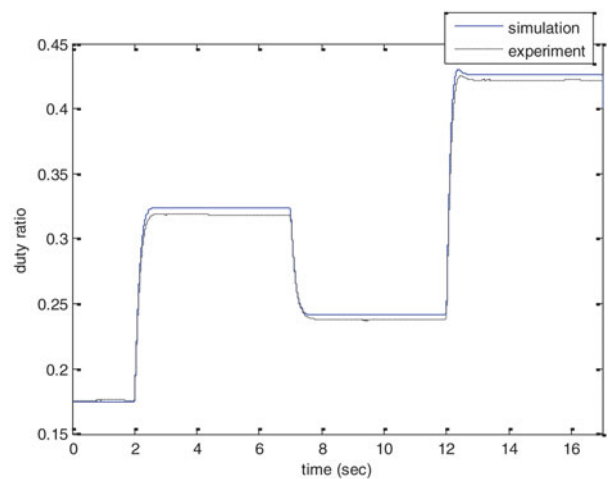
FIGURE 6. Time response of the proposed control scheme and stability analysis: (a) motor speed response, (b) duty ratio response, and (c) controller convergence in the p - q plane.



(a)



(b)



(c)

FIGURE 7. Simulation and experimental responses of the proposed controller under changes of the reference signal: (a) motor speed response, (b) armature voltage response, and (c) duty ratio response.

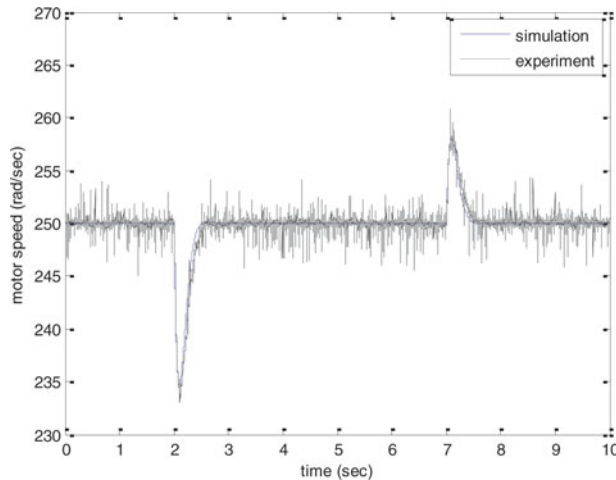


FIGURE 8. Simulation and experimental responses of the proposed control scheme under various changes of the load torque.

In all cases, a smooth response is observed without oscillations or large overshoots in both the system output and the controller signal; indeed, it verifies a low effort and effective controller performance.

6. CONCLUSIONS

A non-linear duty ratio dynamic scheme suitable for speed control applications of a DC motor driven by a DC-DC boost converter has been analyzed and implemented. This controller is based on a simple dynamic performance, which is of the direct speed regulation type and is thus independent from the system parameters while, as proven in the article, guarantees stability and convergence to the desired equilibrium. The theoretical analysis is absolutely verified by simulation and experimental results. The simulations provide an excellent evaluation of the controller performance in both step changes of the speed reference signal and load torque disturbances. Furthermore, the experimental implementation of the proposed controller on a DC-boost converter/DC-motor system provides real-world results, exactly as they are expected from the theoretical analysis and the simulations conducted.

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APPENDIX

$$\tilde{M} = \begin{bmatrix} L & 0 & 0 & 0 & 0 & 0 \\ 0 & C & 0 & 0 & 0 & 0 \\ 0 & 0 & L_a & 0 & 0 & 0 \\ 0 & 0 & 0 & J & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}, \quad \tilde{R} = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & R_a & 0 & 0 & 0 \\ 0 & 0 & 0 & b & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix},$$

$$\tilde{J} = \begin{bmatrix} 0 & -(1-p^2) & 0 & 0 & 0 & 0 \\ 1-p^2 & 0 & -1 & 0 & 0 & 0 \\ 0 & 1 & 0 & -K_e & 0 & 0 \\ 0 & 0 & K_e & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & k(\omega^* - \omega) \\ 0 & 0 & 0 & 0 & -k(\omega^* - \omega) & 0 \end{bmatrix}.$$

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